

# COVID-19 Healthcare Analysis

Comprehensive Healthcare Data Analysis Report

Project 4: Healthcare Domain Analysis

Report Date: September 08, 2026

Dataset: 365 Days, 5 Regions, 1,825 Records

## Executive Summary

This report analyzes COVID-19 healthcare data across five regions over 365 days in 2023 to identify case patterns, mortality differences, vaccination trends, and relationships between reported health indicators.

### VERIFIED ANNUAL STATISTICS:

- Total Cases: 391,062
- Total Deaths: 4,505
- Case Fatality Rate: 1.15%
- Recovery Rate: 86.23%
- Average Vaccination Rate: 47.98%

### KEY FINDINGS:

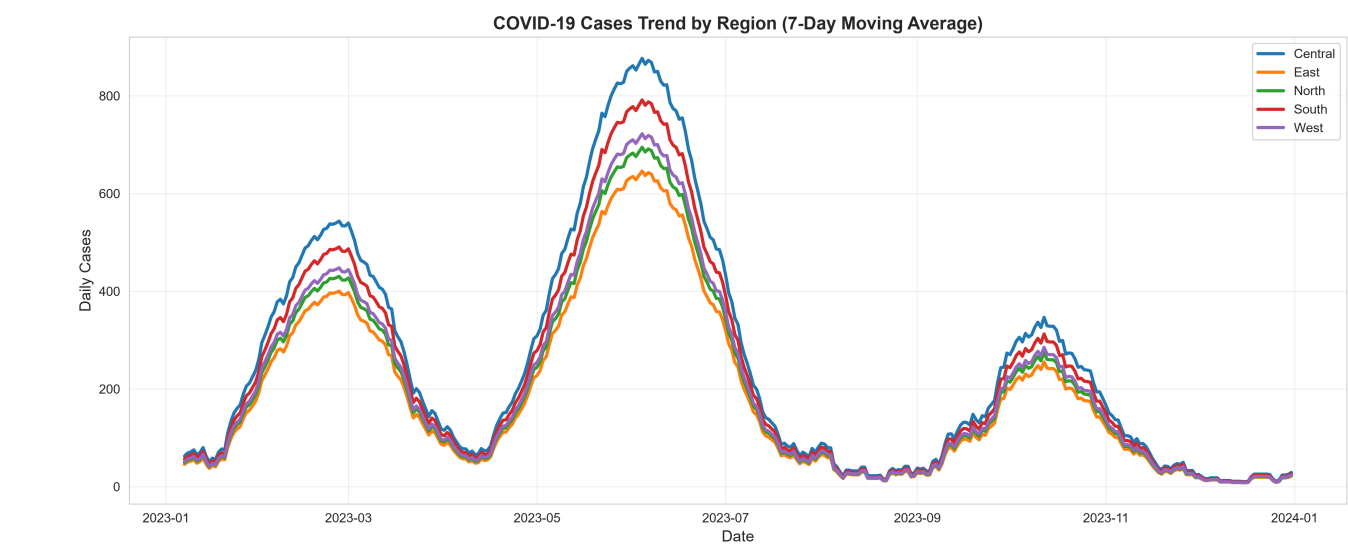
- Several periods of elevated case activity were observed during the year, with the highest monthly total occurring in May at 87,244 cases.
- Cases were 56.7% lower during the high-vaccination phase (60%+) compared with the low-vaccination phase (0-30%).
- Central recorded the highest regional deaths at 1,118, which was 42.1% higher than East.
- Vaccination rate showed a moderate negative correlation with cases (-0.357) and a weaker negative correlation with deaths (-0.160).

### IMPORTANT INTERPRETATION:

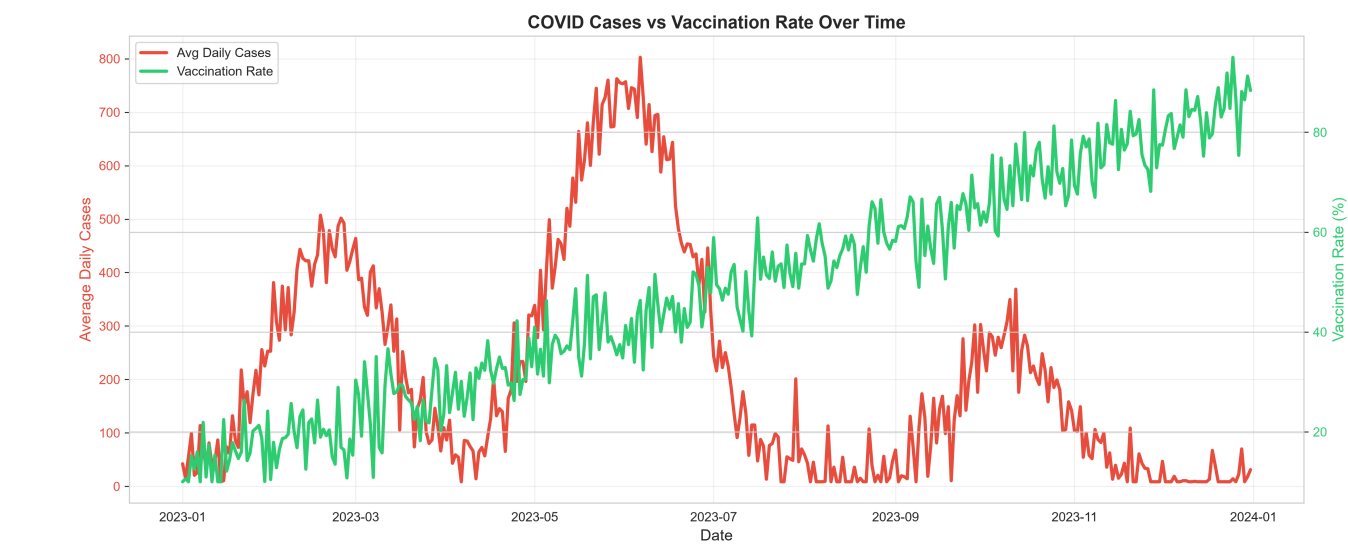
The vaccination findings describe observed associations within the supplied dataset. Correlation and group comparisons alone cannot establish that vaccination directly caused changes in cases or deaths.

# Pandemic Trends by Region

## COVID Cases by Region

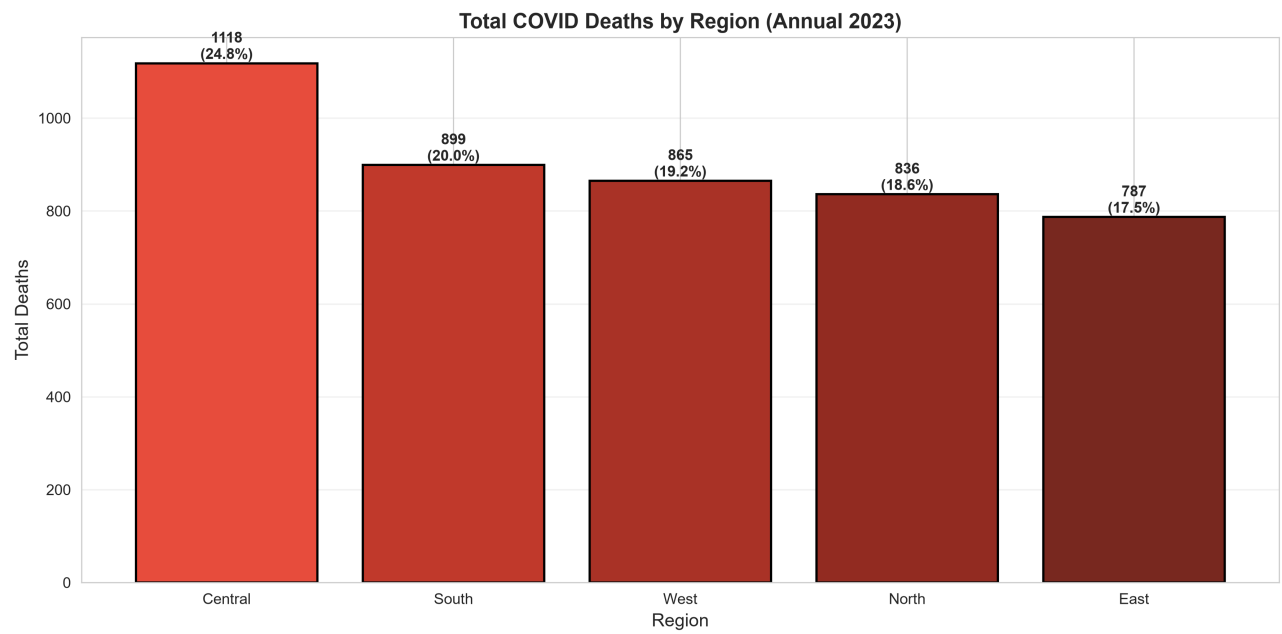


## Vaccination Rate vs Cases Over Time

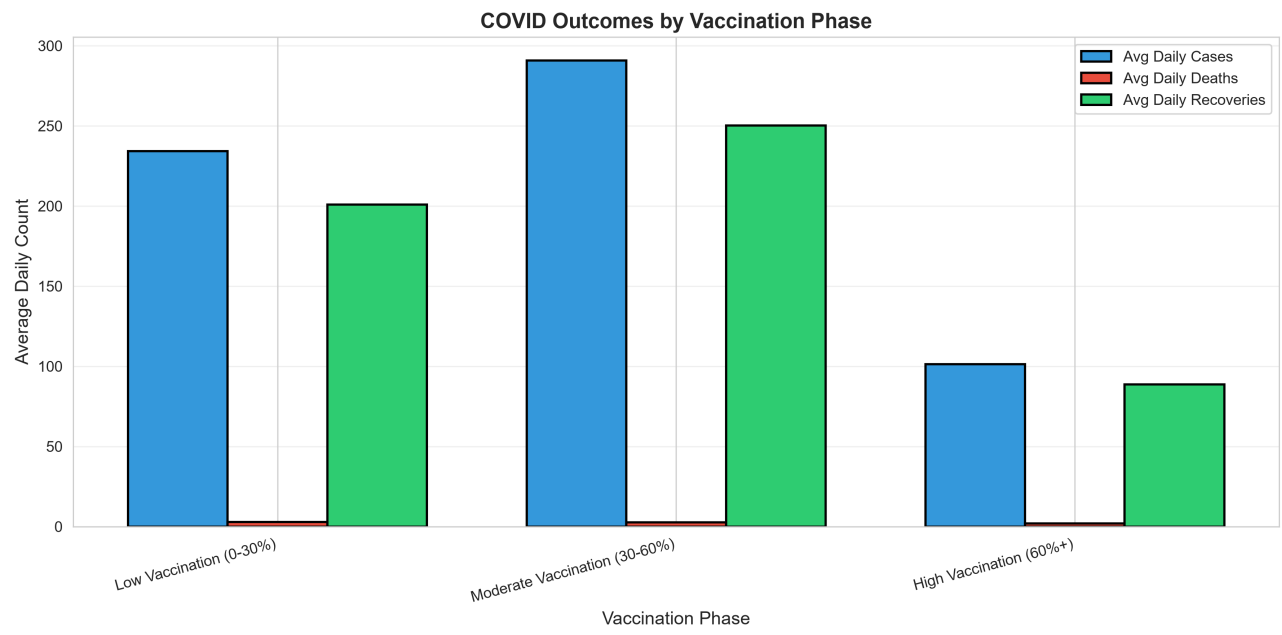


# Regional Mortality and Vaccination Analysis

## Total Deaths by Region



## COVID Outcomes by Vaccination Phase



# Key Findings

## 1. PERIODS OF ELEVATED CASE ACTIVITY

The dataset contains several periods of elevated case activity. Monthly cases reached 57,108 in February, 87,244 in May, and 35,377 in October. May recorded the highest monthly case total in the dataset.

## 2. OBSERVED ASSOCIATION BETWEEN VACCINATION AND CASES

The high-vaccination phase (60%+) recorded 56.7% fewer cases than the low-vaccination phase (0-30%). Vaccination rate also showed a moderate negative correlation with daily cases (-0.357). These findings represent observed associations and do not establish causation.

## 3. CENTRAL REGION MORTALITY DISPARITY

Central recorded the highest number of deaths at 1,118, while East recorded 787. Central therefore recorded 42.1% more deaths than East. This difference warrants further investigation using additional regional healthcare and population-level data.

## 4. VACCINATION-DEATH ASSOCIATION WAS WEAKER THAN CASE ASSOCIATION

The correlation between vaccination rate and deaths was approximately -0.160, compared with -0.357 for vaccination and cases. This indicates a weaker linear association between vaccination rate and daily deaths in this dataset.

## 5. CASES AND RECOVERIES SHOWED A STRONG LINEAR RELATIONSHIP

Daily cases and recoveries showed a correlation of approximately 0.996, while cases and active cases showed a correlation of approximately 1.000. These relationships are expected to be influenced by how the reported healthcare indicators are constructed and should be interpreted as statistical associations.

## 6. REGIONAL PATTERNS SHOW PERIODIC SIMILARITIES

The regions display periods of increased and decreased case activity during the year. However, the available analysis does not establish that all regions followed identical patterns or that regional changes were caused by the same factors.

# Recommendations

## IMMEDIATE PRIORITIES:

1. Investigate the mortality difference observed in the Central region.
2. Establish a monitoring dashboard for cases, deaths, recoveries, active cases, and vaccination rates.
3. Monitor periods of elevated case activity to support healthcare resource planning.
4. Review regional healthcare capacity and reporting consistency.

## MEDIUM-TERM PRIORITIES:

1. Evaluate vaccination coverage alongside additional factors such as age distribution, healthcare access, population density, and circulating variants where data is available.
2. Strengthen regional surveillance and reporting systems.
3. Use observed vaccination and case patterns as inputs for public-health planning rather than treating them as proof of causation.
4. Develop targeted health communication based on regional needs.

## LONG-TERM PRIORITIES:

1. Develop surge-capacity protocols for periods of increased healthcare demand.
2. Improve pandemic preparedness and regional healthcare equity.
3. Build forecasting models using longer historical datasets and additional explanatory variables.
4. Conduct longitudinal analysis to evaluate whether observed relationships persist over time.

## PUBLIC HEALTH MONITORING PRIORITIES:

- Monitor vaccination coverage, including periods above and below the 60% phase used in this analysis.
- Track regional mortality differences.
- Strengthen early-warning surveillance.
- Evaluate healthcare access and capacity differences between regions.
- Compare future observations against the 2023 dataset baseline.

The recommendations above are planning considerations derived from the observed dataset. They do not represent guaranteed intervention outcomes.

# Data Interpretation & Limitations

This analysis is based on 365 days of observational healthcare data across five regions. The results describe patterns within the supplied dataset.

## 1. CORRELATION DOES NOT ESTABLISH CAUSATION

Negative correlations between vaccination and cases or deaths do not prove that changes in vaccination directly caused changes in those outcomes.

## 2. OBSERVED GROUP DIFFERENCES ARE NOT CAUSAL EFFECTS

The 56.7% difference in cases between high- and low-vaccination phases is a descriptive comparison. Other time-dependent or regional factors may contribute to the observed difference.

## 3. LIMITED CONTEXTUAL VARIABLES

The dataset does not provide sufficient information about factors such as population age structure, healthcare capacity, population density, variants, mobility, testing practices, or individual vaccination status.

## 4. REGIONAL DEATH COUNTS

Differences in total deaths between regions may reflect population size and other structural differences. Death counts should therefore not be interpreted as direct measures of healthcare quality without additional normalization and contextual data.

## 5. ONE-YEAR OBSERVATION PERIOD

The analysis covers a single year and should not be used by itself to establish long-term pandemic trends or recurring seasonal patterns.

## 6. PREDICTIVE LIMITATIONS

Historical relationships in this dataset should not be treated as guaranteed predictors of future cases, deaths, or vaccination outcomes.

Future analysis could incorporate population-normalized rates, age-specific data, healthcare capacity, variant information, testing volume, mobility, and longer time-series observations.

## Conclusion

The 2023 healthcare dataset shows several periods of elevated COVID-19 case activity across five regions, with May recording the highest monthly total of 87,244 cases. Overall, the dataset contains 391,062 reported cases and 4,505 deaths, corresponding to a case fatality rate of 1.15%.

A notable observed association exists between vaccination phase and reported cases. The high-vaccination phase (60%+) recorded 56.7% fewer cases than the low-vaccination phase (0-30%), while vaccination rate showed a correlation of -0.357 with daily cases. However, these findings are observational and cannot establish a causal effect of vaccination from this dataset alone.

Central recorded the highest regional death count at 1,118, which was 42.1% higher than East. This regional difference highlights the value of investigating healthcare capacity, population characteristics, and other contextual factors.

Overall, the analysis provides a quantitative overview of COVID-19 case, mortality, recovery, and vaccination patterns in the supplied dataset. It can serve as a baseline for further healthcare analytics while recognizing the limitations of one-year observational data and correlation-based analysis.